باسمه تعالی

آزمایشگاه FPGA

پیش گزارش 1

کیاوش ربانی سودابی - 9821663



سوال 1-

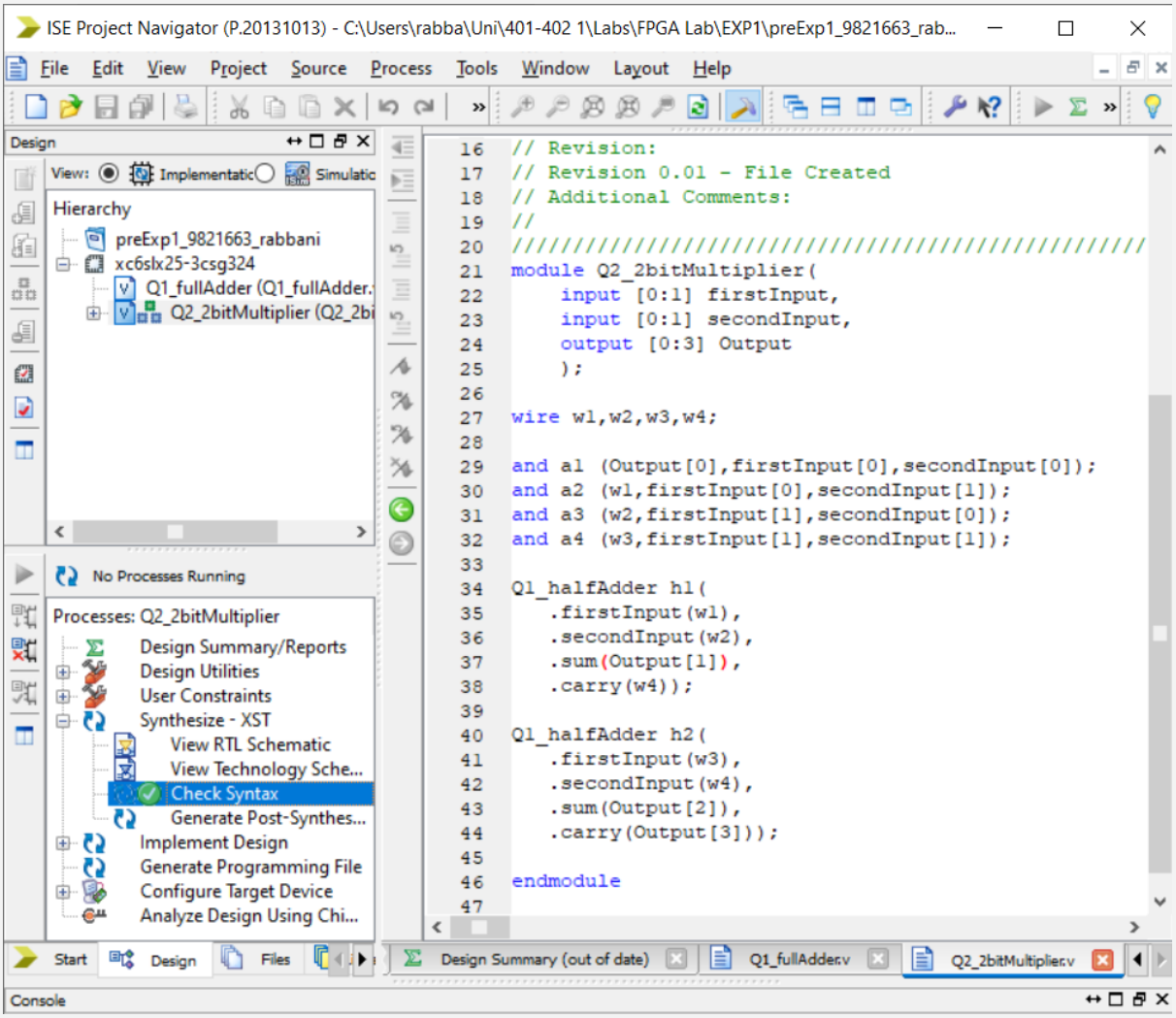
Half Adder:

```
20  //////////////////////////////////////////////////////////////////
21  module Q1_halfAdder(
22      input firstInput,
23      input secondInput,
24      output sum,
25      output carry
26      );
27
28  xor x (sum,firstInput,secondInput);
29  and a (carry,fristInput,secondInput);
30
31  endmodule
32
```

Full Adder:

```
20  //////////////////////////////////////////////////////////////////
21  module Q1_fullAdder(
22      input firstInput,
23      input secondInput,
24      input carryIn,
25      output sum,
26      output carryOut
27      );
28
29  wire w1,w2,w3;
30
31  xor x1 (w1,firstInput,secondInput);
32  xor x2 (sum,carryIn,w1);
33  and a1 (w2,w1,carryIn);
34  and a2 (w3,firstInput,secondInput);
35  or o (carryOut,w2,w3);
36
37  endmodule
38
```

سوال 2-


ISE Project Navigator (P.20131013) - C:\Users\rabba\Uni\401-402 1\Labs\FPGA Lab\EXP1\preExp1_9821663_rab...
File Edit View Project Source Process Tools Window Layout Help
Design
View: Implementatic Simulatic
Hierarchy
preExp1_9821663_rabbani
xc6slx25-3csg324
Q1_fullAdder (Q1_fullAdder.
Q2_2bitMultiplier (Q2_2bi
No Processes Running
Processes: Q2_2bitMultiplier
Design Summary/Reports
Design Utilities
User Constraints
Synthesize - XST
View RTL Schematic
View Technology Sche...
Check Syntax
Generate Post-Synthes...
Implement Design
Generate Programming File
Configure Target Device
Analyze Design Using Chi...
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 //
21 module Q2_2bitMultiplier(
22 input [0:1] firstInput,
23 input [0:1] secondInput,
24 output [0:3] Output
25);
26
27 wire w1,w2,w3,w4;
28
29 and a1 (Output[0],firstInput[0],secondInput[0]);
30 and a2 (w1,firstInput[0],secondInput[1]);
31 and a3 (w2,firstInput[1],secondInput[0]);
32 and a4 (w3,firstInput[1],secondInput[1]);
33
34 Q1_halfAdder h1(
35 .firstInput(w1),
36 .secondInput(w2),
37 .sum(Output[1]),
38 .carry(w4));
39
40 Q1_halfAdder h2(
41 .firstInput(w3),
42 .secondInput(w4),
43 .sum(Output[2]),
44 .carry(Output[3]));
45
46 endmodule
47
Start Design Files
Design Summary (out of date)
Q1_fullAdder.v
Q2_2bitMultiplier.v
Console

سوال 3-

ناقص!!!

```
21  module Q3_4bitMultiplier(
22                      input [3:0] a,
23                      input [3:0] b,
24                      output reg [7:0] out);
25
26  wire [3:0] mul1,mul2,mul3,mul4;
27  wire [3:0] shMul2,shMul3,shMul4;
28  wire [3:0] sum1,sum2;
29
30
31  Q2_2bitMultiplier m1(
32      .firstInput(a[1:0]),
33      .secondInput(b[1:0]),
34      .Output(mul1));
35
36  Q2_2bitMultiplier m2(
37      .firstInput(a[1:0]),
38      .secondInput(b[3:2]),
39      .Output(mul2));
40
41  Q2_2bitMultiplier m3(
42      .firstInput(a[3:2]),
43      .secondInput(b[1:0]),
44      .Output(mul3));
45
46  Q2_2bitMultiplier m4(
47      .firstInput(a[3:2]),
48      .secondInput(b[3:2]),
49      .Output(mul4));
50
51  assign shMul2 = mul2 << 2;
52  assign shMul3 = mul3 << 2;
53  assign shMul4 = mul4 << 4;
54
55  Q3_4bitFulladder A1(
56      .a)
57
```

Q1.v | Design Summary (out of date)